CURRICULUM VITAE

Joaquin Ramirez-Medina

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Education

MASc, Mechanical Engineering Sep 2024 to present

York University

- Surface Engineering and Instrumentation Lab (SEiL), supervised by Prof. Alidad Amirfazli
- Thesis on electrowetting-enhanced dropwise condensation and LLM-agent tools for research
- Coursework GPA 8.7 / 9.0

BEng, Mechanical Engineering Sep 2020 to Jun 2024

York University

- First Class Standing, Dean's Honour Roll

BSc, Biology Sep 2013 to Apr 2018

York University

Research experience

Graduate researcher and teaching assistant Sep 2024 to present

SEiL, York University

- Electrowetting control of dropwise condensation, from electrode design and cleanroom fabrication to control software and wind-tunnel experiments
- ARIA, a multi-LLM system for literature review and research planning

Research intern Nov 2025 to Dec 2025

Institut de Physique de Nice (INPHYNI), Université Côte d'Azur, France

- Python and OpenCV method for contact angles of vibrating droplets from high-speed imaging
- Invited seminar on the electrowetting condensation work

Undergraduate research associate (NSERC USRA) May 2023 to Aug 2024

SEiL, York University

- Led the preliminary version of ARIA
- Co-developed Prof Notes AI, an LLM-agent course Q&A tool

Undergraduate research associate Jul 2021 to May 2023

ACuTE microfluidics lab, York University

- Acoustofluidic rotation of Drosophila embryos in a microfluidic device
- Microfluidic cardiac assays and MATLAB analysis for SARS-CoV-2 ORF3a studies in Drosophila larvae
- Acoustofluidic focusing of microparticles in a glass capillary

Publications

Electrowetting for Multi-Scale Droplet Coalescence and Transport in Dropwise Condensation Aug 2026

J. Ramirez-Medina, A. Amirfazli. Proceedings of the 18th International Heat Transfer Conference (IHTC-18), Rio de Janeiro, Brazil.

AI Enabled Automated Student Q&A Tool: Prof Notes AI Jul 2025

A. Asim, J. Ramirez-Medina, A. Amirfazli. Proceedings of the Canadian Engineering Education Association (CEEA).

Accelerating Scientific Research Through a Multi-LLM Framework Feb 2025

J. Ramirez-Medina, M. Ataei, A. Amirfazli. arXiv preprint arXiv:2502.07960.

Acoustofluidic Rotational Manipulation of Drosophila Embryos in a Microfluidic Device 2022

A. Zabihihesari, J. Ramirez-Medina, P. Rezai. IEEE EMBS Micro and Nanotechnology in Medicine (MNM), Hawaii, USA.

Talks and posters

Talk	Electrowetting for multi-scale droplet coalescence and transport in dropwise condensation (oral and poster) 18th International Heat Transfer Conference (IHTC-18), Rio de Janeiro, Brazil	Aug 2026
Talk	Invited seminar on electrowetting-enhanced dropwise condensation Institut de Physique de Nice (INPHYNI), Université Côte d'Azur, Nice, France	Dec 2025
Talk	Application of AI to Accelerate Research Processes: Case of Dropwise Condensation 99th ACS Colloids and Surface Science Symposium and 18th IACIS Conference, Edmonton	Jun 2025
Poster	Prof Notes AI Lassonde Undergraduate Summer Research Conference, York University	Aug 2024
Poster	AI for research, applied to dropwise condensation Lassonde Undergraduate Summer Research Conference, York University	Aug 2023
Talk	Cardiac dysfunction in Drosophila larvae expressing SARS-CoV-2 ORF3a Lassonde Undergraduate Summer Research Conference, York University	Aug 2022

Teaching

Teaching assistant, Lassonde School of Engineering Sep 2024 to Apr 2026

MECH 2112	Mechanical Engineering, Professionalism and Society	Winter 2026
MECH 4402	Simulation Tools for Design and Analysis	Fall 2025
MECH 4203	Energy Conversion and Storage	Winter 2025
MECH 4510	Advanced Mechanical Technologies	Winter 2025

MECH 4403	Artificial Intelligence and Generative Design for Mechanical Systems	Fall 2024
MECH 4502	Vibrations	Fall 2024

Mentoring

- LURA student in Mechanical Engineering, numerical methods (May to Aug 2025)
- Mechanical Engineering intern from ENSTA (France), image processing for droplet motion and coalescence tracking
- Software Engineering student (2024 to 2025), work published as a peer-reviewed CEEA paper

Awards

Aug 2026	Academic Excellence Fund Faculty of Graduate Studies, York University
Jul 2026	Mechanical Engineering Excellence Award for Graduate Research Department of Mechanical Engineering, York University
2026	York International Mobility Award York International, York University
May 2023 and May 2024	NSERC Undergraduate Student Research Award (USRA) NSERC
May 2022 and Oct 2022	Lassonde Undergraduate Research Award (LURA) Lassonde School of Engineering
2024	First Class Standing, BEng York University
2020 to 2024	Dean's Honour Roll Lassonde School of Engineering

Skills

Microfabrication	Photolithography, e-beam evaporation, lift-off, PECVD, mask aligner, spin coating
Experiments	High-speed imaging, closed-loop humidity and temperature control, wind-tunnel testing
Electronics	PCB design (KiCad), microcontrollers, solid-state relay switching, soldering
Software	Python, OpenCV, MATLAB, SQL, LlamaIndex, Docker, Google Cloud, Firebase
CAD and analysis	SolidWorks, AutoCAD, ImageJ, Tracker
Languages	English, Spanish